



Bridging the gap between rubbers and plastics: a review on thermoplastic polyolefin elastomers

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Abstract

Thermoplastic elastomers (TPE) are studied for their unique properties of being easily processes and recycled. This paper focuses on the wide scope of thermoplastic polyolefin elastomers (TPO/TPE-O) which is one of the major classes of TPE. The performance of TPO is the cumulative effect of polyolefins and rubbers with the ease of processing due to the presence of thermoplastics. The study focuses on the classification of TPO, their routes of synthesis, composites, applications, and its impact on environment. The mechanical blending of polyolefins and conventional rubber was carried out to manufacture TPO. In further developments, the polyolefin elastomers (POE) were synthesized by copolymerization techniques to overcome the drawback of mechanical blending techniques. Environment-friendly blends of TPO are synthesized using industrial and municipal waste which includes scrap rubber tires, marble waste-filled polypropylene, computer body waste, etc. This technology will help in reducing the issue of landfills. The composites of TPO have been studied using different types of fillers which may be synthetic or bio-based. These include talc, carbon black, carbon nanotubes (CNT), Kenaf fiber, pineapple leaf fiber, etc. TPO have found their use in encapsulation, in electrical insulation, roofing, medical devices, and the automobile industry. Also, impact of using TPOs on environment is discussed here qualitatively.

Keywords Thermoplastic elastomers · Thermoplastic polyolefin elastomer · Medical · Automobile · Rubber

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Introduction

Thermoplastic polymers possess excellent properties such as low density, high chemical resistance, and heat resistance and can be fabricated using common processing techniques [1–3]. On the other hand, rubbers are used for their elastomeric properties such as good tear strength, impact resistance, and resilience. When these two materials are blended or mixed, they give a synergistic effect. These materials are called as thermoplastic elastomers (TPEs). Thermoplastic elastomers (TPEs) are considered a commercially fascinating class of polymers. These polymers are composed of more than one phase, which consists of: (i) semicrystalline thermoplastics and (ii) elastomers phase [4–6]. TPEs have properties like thermoset rubbers but have processing flexibility like thermoplastics, which provides a reduction in cost and better design opportunities [7]. There are different types of TPEs manufactured and used commercially that are represented in Fig. 1 [8, 9].

Conventional rubbers are difficult to process and they require high processing time. As they are chemically cross-linked polymers, recycling is the major issue that leads to increased waste in landfills [10–12]. Hence, the TPE bridge between thermoset rubbers and thermoplastic polyolefins and have optimum processing and performance properties. This paper reports the technological and commercial importance of studying thermoplastic polyolefin elastomer (TPO). A mechanical blending of polyolefins with various elastomers such as EPDM, ethylene-propylene rubber (EPR), and ethylene-octene copolymers (EOC) results in blend type-TPOs. The vulcanization of these elastomeric compounds had also been carried with olefins [8, 13–17]. Products of copolymerization of olefin were later used in blending with thermoplastic polyolefins which depicted better results. TPOs are further compounded to obtain increment in certain properties such as electrical, thermal for specific applications [7, 17]. It is reported that TPOs are processed using common processing techniques such as injection molding, extrusion, and thermoforming [8, 19]. TPOs were formulated using waste rubbers and polyolefins which reduced stress on the environment by reducing plastic waste. TPOs have found their application in medical, automobile, microwave, encapsulation, and telecommunication [18–21].

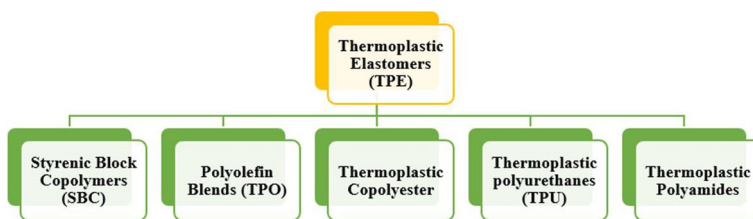


Fig. 1 Types of thermoplastic elastomers

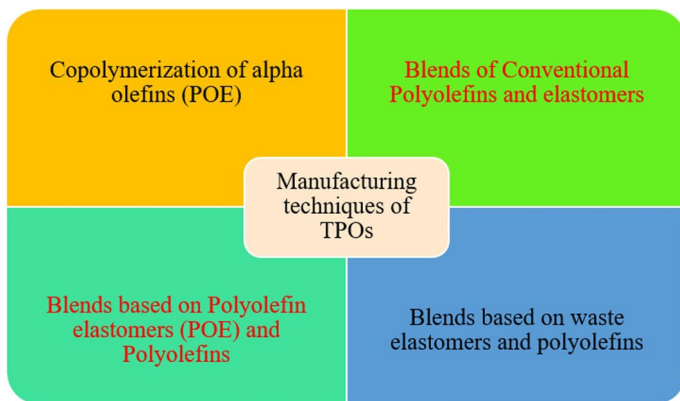


Fig. 2 Manufacturing techniques of TPO

Table 1 Few important mechanical properties of conventional elastomers [22]

Sr. no	Conventional rubbers	Tensile strength (MPa)	% Elongation
1	Natural rubber	4–25	100–600
2	Styrene-butadiene rubber	4–25	100–500
3	Nitrile-butadiene rubber	4–18	100–400
4	Chloroprene rubber	4–20	100–500
5	Silicone rubbers	3–10	100–400
6	EPDM	4–18	100–400
7	Polyurethane rubbers	15–30	100–800

In this paper, we have discussed the classification of TPOs, its mode of synthesis which includes mechanical blending and copolymerization, compounding of TPOs with different fillers, its application in various sectors, and its sustainability.

Methods of manufacturing

Different routes of manufacturing thermoplastic polyolefin elastomers are shown in Fig. 2 and explained below.

Blending of polymer–polymer, polymer–rubber was carried out to improve some or the other required properties of blend during application. Conventional rubbers show good tensile strength, elongation at break but blending them with olefins further improve their processing properties. Data in Table 1 give us glimpse of important mechanical properties of conventional rubbers/elastomers [22].

From blends

Blends of natural rubber and polyolefins: Natural rubber is one of the most commonly used rubbers to be blend with polyolefins for the synthesis of TPOs. Study reported that blending of linear-low-density polyethylene (LLDPE) with 1% liquid natural rubber (LNR), 50% natural rubber had resulted in TPOs with homogeneity. The presence of LNR in the blend was responsible for homogenous nature of blend. Figure 3 shows the effect of addition of LNR in NR/LLDPE blend [23]. Dynamically vulcanized ultra-low-density polyethylene (ULDPE)/natural rubber (NR) blends were prepared in different mixing ratios along with a cross-linking agent. Blends containing 50–60 wt% of NR have shown to have co-continuous morphology. Cross-linking agents in 40–50 wt% of ULDPE along with NR have shown best results with improved abrasion and aging resistance. The new blend of NR/ULDPE was compared with NR/SBR blend for its tensile strength and elongation at break. It was safely concluded due to synergistic effect of NR/ULDPE, and it has better tensile properties than NR/SBR blend. The reason was assumed to lie in microstructure of blend and also non-polarity of both the constituents. Figure 4 shows SEM analysis of NR/ULDPE blends explaining its morphology and hence its components compatibility [24]. To overcome the drawback associated with the blending of NR with isotactic polypropylene (iPP) which included thermal degradation and incompatibility due to the higher molecular weight of NR, the mastication of natural rubber was carried before blending from about 0–20 minutes and its molecular weight (M_n) was determined with GPC. With about 10–15 minutes of mastication, both better processing and tensile properties of NR/PP were obtained. Curing systems of dicumyl peroxide are shown to have better results. [25]. The blend of natural rubber and PP was dynamically vulcanized and compression-molded. The samples were studied for morphological behavior of these blends, which revealed that un-vulcanized blends

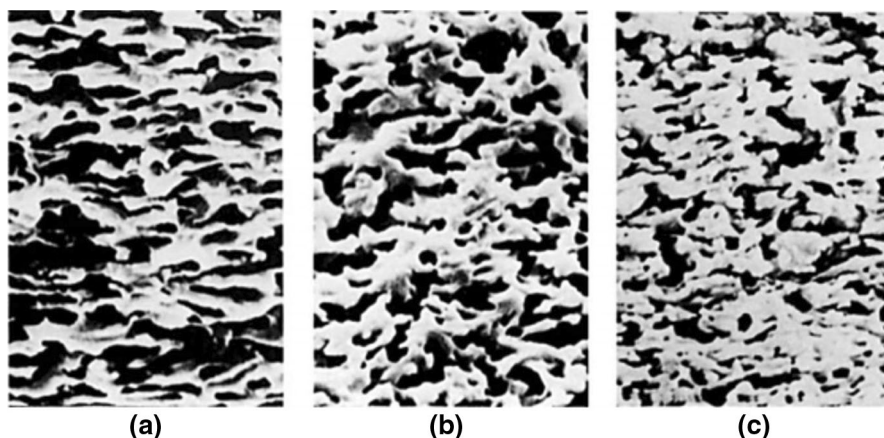
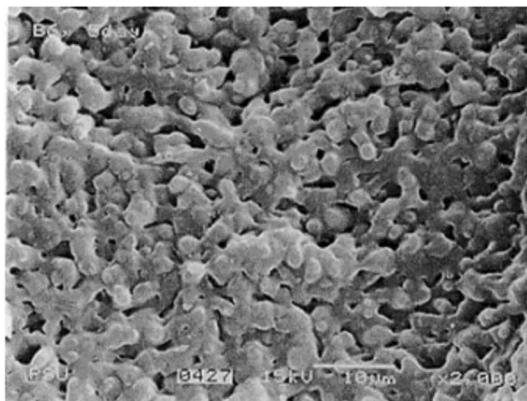
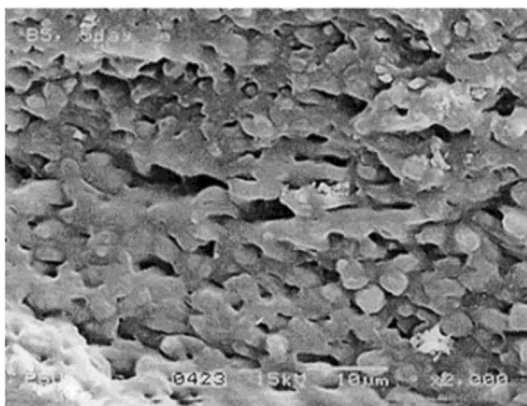


Fig. 3 SEM analysis of toluene extracted 50:50 NR/LLDPE blends with **a** 0% LNR, **b** 10% LNR, and **c** 20% LNR [Reprinted from Blending of Natural Rubber with Linear Low-Density Polyethylene, IBRAHIM ABDULLAH, SAHRIM AHMAD, CHE SOM SULAIMAN, *Journal of Applied polymer Science*, 58, 1125–1133, 1995, with permission from John Wiley and Sons]

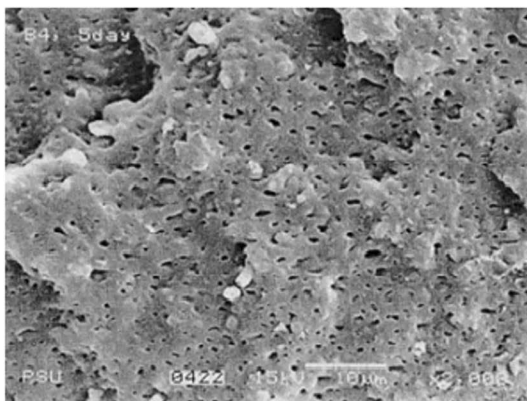
Fig. 4 SEM analysis of NR/ULDPE blends after etching with p-xylene **a** NR/ULDPE (50/50), **b** NR/ULDPE (60/40), and **c** NR/ULDPE (70/30) [Reprinted from Development of Novel Elastomeric Blends Containing Natural Rubber and Ultra-Low-Density polyethylene, V. TANRATTANAKULA, W. UDOMKICHDECHA, Journal of Applied polymer Science, 82. 650–660, 2001, with permission from John Wiley and Sons]



(a)



(b)



(c)

have co-continuous morphology and vulcanized blends have dispersed rubber in the olefin matrix. Dynamic vulcanization thus showed improved tensile strength, elongation at break, and reported no increment in hardness and tear strength as such [26]. Blends of epoxidized natural rubber (ENR) and polypropylene (PP) were prepared by dynamic vulcanization using sulfur and studied. It was reported that with the increase in ENR content, it led to improved toughness and flexibility, with optimum 40/60 PP/ENR ratio it showed improvement in elongation at break and impact strength but decrease in tensile strength. It was stated that as rubber content increased, there was restriction in formation of crystalline domains, which in turn reduced tensile strength. The cross-linked PP/ENR blend showed improved elastic behavior but decreased Young's Modulus [27].

Blends of other conventional polymers with polyolefins: Blends based on ethylene vinyl acetate (EVA) and PP had shown advantages such as low cost, low density, and ease of processing. Dynamic mechanical spectroscopy revealed there exist two separate phases of PP and EVA, indicating the immiscibility of two phases. But it was reported that some amount of compatibility was obtained with cross-linking. An increase in EVA content decreased storage modulus but improved damping properties [28]. A unique compatibilized blend of PP/EPDM was prepared and studied for its improved mechanical properties. A triblock copolymer from PP-g-MA, EPDM-g-MA, and PA-66 was prepared which acted as compatibilizer for blend. With 0.5 wt% use of compatibilizer, there found improvement in low temperature impact strength [29]. A comparative study based on PP/EPDM composite and PP/NR composite was done. The filler used was Vetiver grass, and its surface was chemical treated with NaOH and was used to improve mechanical properties. Up to 20 wt% of rubber, composites had significant increase in elongation at break and impact strength. PP/EPDM composites had slightly higher impact and tensile strength than PP/NR blend [30, 31]. Blends based on high flow PP and branched EPDM were studied. The study explained that, for impact toughening of PP, long-chain branched EPDM modifiers were better than linear Ethylene-propylene copolymers [32]. The major drawback of isotactic PP reported is its high T_g which lead to rigidity and brittleness at low temperatures. Also, PP being nonpolar, incorporation of plasticizers was not effective. Thus, grafting of PP with maleic anhydride and unsaturated carboxyl esters was carried out earlier. Later study was made where PP was modified with itaconic acid derivatives, including particularly monomethyl itaconate (MMI) and dimethyl itaconate (DMI) by using 2,5-dimethyl-2,5-bis(tert-butylperoxy) hexane. Blends of PP and EPDM made were of great interest in automobile. Thus, PP grafted with DMI shown to have better processability and adhesion [33]. A blend comprising of maleated ethylene-propylene monomer (EPM) and PP was dynamically vulcanized using dicumyl peroxide (DCP) as cross-linking agent, triallyl cyanurate as co-agent and maleated PP as compatibilizer. It was depicted with varying quantity of DCP, properties of blends were affected due to change in crystallinity. Optimum concentration of DCP obtained was about 2 phr. Blend obtained was shown to have excellent recycling and aging properties [34]. Blends based on acrylic rubber and PP were synthesized by compatibilization and dynamic vulcanization. The addition of maleic

anhydride-grafted PP (MA-g-PP) and triethylene tetramine (TETA) as compatibilizer resulted in the improvement in mechanical properties and oil resistance. With vulcanization, ultimate tensile strength was found to be increased [35]. Blends containing PP and carboxylated nitrile rubber (XNBR) were synthesized by dynamic vulcanization using dicumyl peroxide along with *N,N*-m-phenylene-bis-maleimide (BMI) as a cross-linking agent. PP was functionalized using maleic anhydride and glycidyl methacrylate with triethylene tetramine (TETA). Glycidyl methacrylate functionalized PP acted as efficient compatibilizing agent because of carboxyl groups of XNBR reacting with epoxy groups on PP. It was observed that there was an improvement in ultimate tensile strength and elongation at break, along with good oil resistance in vulcanized PP/XNBR blend and good re-processability [36]. Various polyolefins which included LDPE, HDPE, PP, and random ethylene-propylene static copolymer (REP) were blended with butyl rubber (BR) using dicumyl peroxide (DCP) and by varying BR concentration and DCP concentration. Increase in cross-linking density and BR content into blend improved the elongation at break for some polyolefins such as HDPE, PP and REP while for LDPE it decreased [37]. Blends of polyethylene (PE) with silicone rubber were prepared. Among various PEs used, LLDPE showed higher potential to promote the reaction between two phases. When LLDPE and silicone rubber were mixed at high temperature, it led to free radical sites on the backbone of LLDPE which reacted with vinyl groups of silicone rubber, thereby leading to both grafting and cross-linking [38]. Blends based on HDPE/EPDM along with compatibilization of PE-g-MA and dynamic vulcanization using phenolic resin were studied. It was concluded that an increase in tensile strength and tear strength of the blends was attributed to the presence compatibilizing agent and vulcanization agent. As compared to the un-compatibilized HDPE/EPDM blend, compatibilized ones showed higher values of elongation at break, tear strength, and hydrophobicity [39]. Thus, a very detailed study was done on olefin

Table 2 Mechanical properties of blends made from conventional rubbers/elastomers and polyolefins

Sr. no	Blends of rubbers and polyolefins	Tensile strength	% Elongation	References
1	NR/LLDPE/LNR (50:50:10%)	10.6	1020	[23]
2	NR/LLDPE/LNR (60:40:20%)	10.6	1100	[23]
3	NR/40SBR	17.91	772	[24]
4	NR/40ULDPE	23.43	821	[24]
5	NR/40ULDPE/1% DCP	28.29	827	[24]
6	NR/PP/DCP (60:40:1) (NR-3.7Mn *10 ⁵)	13.48	225	[25]
7	NR/PP (Phenolic curing system) (40:60)	17.54	506	[26]
8	ENR/PP (60:40)	<10	>15	[27]
9	EPDM/PP (EPDM branching index 0.7) 30:70	14.3	–	[32]
10	m-EPM/PP/m-PP/DCP (100:60:10:2)	7.3	240	[34]
11	ACM/PP/PP-g-MA/TETA (50:50:5:0.2)	16.5	190	[35]
12	XNBR/PP/PP-g-GMA/DCP (30:70:5:0.3)	21.8	660	[36]
13	EPDM/HDPE/Phenolic resin (60:40:4.8)	19.9	340	[39]
14	EPDM/HDPE/PE-g-MA (60:40:5)	8.7	678	[39]

and rubber blends with compatibilization and additives. Among all, PP/EPDM was found to give more focus, where its properties were improved using various techniques such as compatibilization, filler (nano and micro) incorporation, and dynamic vulcanization. These blends have found their application in electrical insulations, and shoe soles [40]. A quantitative improvement in mechanical properties from all the blends above is shown in Table 2.

Blends with recycled polyolefins: High-performance thermoplastic elastomers were synthesized using recycled HDPE, olefinic EPDM, SBR, and ground tire rubber along with bitumen using dynamic vulcanization. Bitumen functioned as a reclaiming agent for GTR under treatment condition whereas compatibilizer after processing of recycled HDPE/EPDM/GTR blends. TPEs containing GTR pretreated with bitumen showed improved adhesion between GTR particles and surrounding thermoplastics, when GTR is partially vulcanized. Also blends based on EPDM were of more tensile strength and hardness than of SBR [41]. A study where waste PP collected from municipal solid waste was cleaned, chopped and blended with rubber waste from textile cot industry. Rubber waste mostly consisted of PVC and NBR. It was reported that up to 10 wt% of rubber waste, properties like tensile strength, impact strength and flexural strength increased and later found to have decreased. The reason reported for increase in properties at about 10% was attributed to groups formed during environmental weathering of PP. It led to formation of polar groups such as C=O and C-OH which are polar, later these groups interact with polar groups on rubber (PVC/NBR) and forming cross-linked structure [42]. Blends of industrial waste-filled PP and acrylonitrile-butadiene rubber (ABR) were compatibilized with chlorinated PE, which improved the mechanical properties, aging resistance, tensile strength, oil resistance, and modulus. Chlorinated PE helped in the dispersion of rubber in matrix thereby giving improved oil resistance and thermal aging [43]. To improve the properties of recycled PP, it was compounded with various elastomers, which included styrene-ethylene/butylene-styrene block-copolymer (SEBS), styrene-isoprene-styrene (SIS) block-copolymer along with antioxidant (Topanol). Among all the elastomers used, SIS with 30% of styrene content proved to be the best elastomer which showed improved Izod impact strength of blend and tensile strength at break. Altogether all the blends showed better mechanical properties than unmodified PP (RPP) [44]. Blends containing highly filled (i.e., up to 90%) recycled PE/GTR along with compatibilizer (Engage 8180) were synthesized. With increasing GTR content above 50 wt%, hardness, tensile strength, and flexural modulus found to be decreased for the blend. For tensile strength analysis, it was concluded from data that, up to 40 wt% of GTR loading, tensile strength was not found to be affected. The use of compatibilizer is shown to have improved the properties of the blend such as elongation at break. Thus, industrial-based products can be manufactured from recycled plastic and GTR up to 90% [45].

Blends based on waste elastomer: Recycling of tire rubber waste is an important concern for technologist as this waste is major threat for environment [46]. Thus, technologist had come up with different methods to tackle the waste. The study reported that blending of ground tire rubber (GTR) and PP was carried out. In that

study, ground tire rubber (GTR) and ultrasonically devulcanized rubber (DGTR) were incorporated in PP matrix. DGTR was revulcanized and compounded in PP. Phenolic and sulfur curing systems were used. It was stated that maleic anhydride-grafted PP and phenolic system proved to be the best compatibilizer and curing system, respectively. It was also reported that blends had good processing properties along with higher impact strength than unmodified PP. With sulfur cure system and revulcanized GTR, elongation at break was greater than phenolic cure system [47].

GTR was used to partially substitute EPDM in dynamically vulcanized EPDM/acrylic modified HDPE. About 50% of replacement of EPDM can be done easily without hampering original properties [48]. Another study in which GTR was blended with waste PP (50:50) ratio, with *N,N'*- meta-phenylene dimaleimide (HVA-2) and di(tert-butylperoxyisopropyl) benzene (DTBPIB) as curing agents and activator, respectively, and were tested for its mechanical properties. With increase in curing agent, properties like impact strength, elongation at break was found to be increased. Optimum activator concentration for blend was reported to be 0.6 pphp [49]. The blends of devulcanized waste rubber (DWR) DWR/HDPE with varying ratio were prepared, tensile strength shown to decrease with increase in DWR content. Optimum tensile strength was obtained at 30 wt% and 150KGy of irradiation. Elongation at break was drastically reduced in between 30 and 70 wt%, and thus, about 20 wt% of DWR found to have reasonable elongation at break [50]. The study of blending of GTR and LLDPE was reported, stating that incorporation of 61 wt% of GTR can be done without hampering processing conditions. This blend is shown to have moderate elongation at break and tear resistance, and with the addition of styrene-butadiene-styrene (SBS), its mechanical properties were improved [51]. HDPE and GTR blends with and without coupling agent showed inferior properties and along with that due to difficulty in incorporating higher amounts of GTR in blends, therefore a study was made using maleated-PE (MAPE) as matrix. Blends containing high loading of GTR content (50–90%) and MAPE were synthesized. These blends showed better interfacial adhesion than HDPE when used as matrix. Blends incorporated with 50–70% of GTR in MAPE showed very good tensile strength, elongation at break, elastic behavior, and elasticity increased with increasing rubber content. Among all of these variations studied, 70% GTR had optimum properties [52]. To reduce the load on fresh raw materials, blends containing PP, EPDM, and waste-EPDM (W-EPDM) were prepared for replacing some quantity of neat EPDM using dicumyl peroxide. Optimum properties were obtained at 38.5/31.5/30 quantity of R-EPDM/W-EPDM/PP [53]. The study showed that reclaimed rubber can replace natural rubber in thermoplastic polyolefin elastomer. About 50% of natural rubber can be replaced by reclaimed rubber without hampering properties of the blend [54]. A blend comprising of 40:60 ratio of PP/latex rubber was synthesized and had comparable mechanical properties to NR/PP blend. With dynamic cross-linking and the addition of latex, the brittle failure of PP was changed to ductile type [55]. Blends containing waste rubber tire (WRT) and PP with bitumen were synthesized. These blends have shown enhanced elongation at break, thermal stability, and processing flowability [56, 57]. From the above discussion based on recycled rubber and polyolefins blends, some of their mechanical properties are listed in Table 3.

Table 3 Mechanical properties of blends made from recycled rubbers/elastomers and polyolefins

Sr. no	Blends of recycled olefins and rubbers	Tensile strength (MPa)	Elongation at break (%)	References
1	EPDM/HDPE/(GTR/Bitumen) (40:35:25(1/1))	6.1	615	[41]
2	Rubber dust (PVC/NBR) Waste PP Rubber dust/WPP (10:90)	30	–	[42]
3	NBR/WPP/CPE (30:70:5)	45	120	[43]
4	SIS30/RPP/PE-g-MA (10:84:5)	11.70	10.96	[44]
5	GTR/RPE (10:90)	11.5	90	[45]
6	RGTR/PP/phenolic cure system (20:80:12)	20	15	[47]
7	RGTR/PP/sulfur curing (80:20)	20	40	[47]
8	GTR/PP/HVA-2 (50:50:3)	16	35	[49]
9	GTR/MAPE (70:30)	7	431	[52]

Table 4 Mechanical properties of few commercially marketed POEs [59–68]

Sr. no	Polyolefin elastomers (POE) (Copolymers)	Tensile strength (MPa)	Elongation at break (%)	References
1	ENGAGE TM XLT 8677 (Ethylene-Octene Copolymer)	1.21	>1000	[59]
2	Lucene TM LC100	38	660	[61]
3	Lucene TM LC165	2.2	>800	[60]
4	ENGAGE TM 8842	3	>600	[62]
5	Dow engage ^R 8440	5.9	760	[63]
6	HifaxT CA287AHF	13.99	500	[64]
7	Elastocon ^R 2855	6.55	660	[65]
8	RTP 2800B-40D	11.37	700	[66]
9	Teknor Apex Telcar ^R TL-2365A	6.21	650	[67]
10	Mitsui NOTIOT PN-2070	14	>800	[68]

Table 5 Mechanical properties of blends made from polyolefin elastomers (POE) and polyolefins

Sr. no	Blends of polyolefin elastomers and polyolefins	Tensile strength (MPa)	Elongation at break (%)	References
1	GTR/ENG 8003/LLDPE (25:25:50)	7	500	[71]
2	EBC/PP (20:80)	14.3	138	[74]
3	PEB/LDPE (70:30)	26.4	1133	[76]

Blends of ethylene-octene copolymer (EOC) and polyolefins: Following blends were prepared using copolymerized polyolefin elastomers (POE) and conventional polyolefins. These POE have shown to be excellent impact modifiers for PP [58]. Various companies like Dow, DowDupont, and Lyondell Basel have pioneer work in synthesis of different POEs via copolymerization. Table 4 gives the insight of mechanical properties of POEs manufactured by such companies [59–68]. Also, mechanical properties of these POEs when blended with polyolefins are given in Table 5.

Polyethylene copolymers are used as an impact modifier with PP. It showed that polyethylene copolymers with PP blended together showed lower torque values than EPDM/PP blends thereby showing better processing capability. Mechanical properties like tensile strength of POE/PP blend shown not to have much difference [69]. Toughening behavior of different types of PP with ethylene-octene copolymer (C2–C8) was studied and compared with PP hetro-phase. PP hetro-phase is formed in reactor when EPR is dispersed in PP matrix. Impact strength of PP hetro-phase was lower than blended PP variants. PP copolymer blended with 25% C2–C8 content had the highest impact strength [70]. The addition of ethylene-1-octene (EOC) to LLDPE/GTR (Ground Tire Rubber) blend improved mechanical properties. With 5% addition of EOC in LLDPE/PP (50/50) blend shown a significant increase in ultimate tensile strength. Thus, EOC acted as compatibilizing agent for LLDPE/GTR blend. With the addition of 30% of EOC to LLDPE/GTR blend the T_g of LLDPE and GTR shown to have approached the closest. Thus, the comparative study was made using different EOCs and their mechanical properties were reported. Highest tensile strength among all achieved was about 6–7MPa. Thus, optimum mechanical properties were reported for LLDPE/ENG8003/GTR in ratio 50/25/25 [71]. Reactive compatibility is established between ethylene-octene copolymers and PP blend by incorporating reactive compounds capable of reacting at the interface. Thus, amine functionalized groups were introduced into PP in solution reaction. Now amine functionalized PP was not made available easily, thus PP-MA was used and then modified and used as compatibilizer [72]. Poly(4-methyl-1-pentene) (PMP) was toughened using various impact modifiers such as EPDM, POE (EOC-ENGAGE™ 8150), and SBS. POE/PMP blend showed good interfacial compatibility also good impact strength due to better dispersion of rubber in PMP matrix. PMP/POE blends shown to have best comprehensive properties [73].

Blends with other polyolefin elastomers Blends of PP with ethylene-butylene-copolymer (EBC) and ethylene-octene copolymer (EOC) were prepared. These blends showed complete immiscibility in melt and solid-state. These blends shown to have improved ductility. Mechanical properties like impact strength, elongation at break, and impact energy were found to be greater for EBC than EOC which was attributed to higher crystallinity [74]. The storage modulus of POE/cyclic olefin copolymers (COC) blend was studied and compared with neat COC and neat POE. In blends with COC as dispersed phase had narrow particle size distribution morphology than POE when dispersed, this was attributed to the lower viscosity of COC than POE. Blends with 60% POE have co-continuous morphology. Storage modulus is increased with POE content and optimum at 70/30 ratio of COC/POE

[75]. Blends based on ethylene-butene copolymers and LDPE with the ratio of 70:30 were studied and shown to have best properties. Ethylene vinyl acetate was used as compatibilizer at lower loadings in this blend. The resultant blend showed high tensile strength and low hardness properties [76]. PP was blended with two types of POEs based on ethylene (TPO1) and propylene (TPO2). With 20 wt% of propylene-based TPO blend, impact strength and toughness properties enhanced and attributed to finer dispersed morphology. For ethylene-based TPO blend, there also reported improvement in impact strength and wide distribution of interparticle distance [77].

Blends compatibilized with copolymers: The following blends were prepared using specific compatibilizers

TPO/PLA blend: Thermoplastic polyolefin elastomer-grafted polylactide (TPO-PLA) was prepared by grafting polylactide on to maleic anhydride functionalized TPO in the presence of 4-dimethylaminopyridine. This compound will then act as compatibilizer for TPO/PLA blend and will improve the compatibility. With compatibilizer concentration up to 2.5%, tensile properties, elongation at break, and toughness increased, but on further addition, properties shown to be decreased. TPO-PLA compatibilizer tends to decrease the interfacial tension between two immiscible polymers thereby improving mechanical properties [78].

TPO/TPU blend: Incorporation of hydrogenated polybutadiene (HPB) segment on to TPU improved the compatibility of TPO/TPU blends. With HPB segment incorporated, the particle size of a dispersed phase in blend decreased. Thus, the blend showed good tensile properties. The segment was shown to lie between two phases thereby reducing interfacial tension and acting as compatibilizer [79].

Copolymerization

The blending of rubber and thermoplastic was reported the most conventional route for the synthesis of thermoplastic elastomer. But due to positive enthalpy and low entropy, this mixing leads to mostly immiscible blends. Thus, the use of compatibilizers in blends reduced the interfacial tension and improved miscibility. Later on, to overcome the drawbacks in blending, copolymerization technique was adopted to synthesize polyolefin elastomers (POE). Different types of catalysts were used to carry out polymerization which included Zeigler–Nata (ZN) and metallocene. Metallocene differed from ZN as they are homogenous catalysts soluble in hydrocarbons, giving more yield than ZN [80, 81]. Commercial copolymers such as ethylene-1-octene, ethylene-1-butene exhibited excellent properties [82].

Engage polyolefin elastomers are C2–C8 copolymers which are produced by a metallocene catalyst. These were produced using INSITE catalyst technology, which had good control over the polymer's structure-property [83].

Thus, various such POEs were synthesized using a catalytic system containing different copolymer contents. POE containing divinylbenzene (DVB) units, along with ethylene, propylene, and octene were synthesized which included poly(ethylene-ter-propylene-ter-divinylbenzene) (EP-DVB) and poly(ethylene-ter-1-octene-ter-divinylbenzene) (EO-DVB). The catalyst used was Et(Ind)₂ZrCl₂/

MAO. Good solubility in organic solvents, controlled DVB content, higher molecular weight, and narrow molecular weight distribution was obtained for copolymers [84]. Random branched linear polyethylene (RHBLPE) was synthesized using one-step polymerization of ethylene. The catalyst used was 2,6-dibenzhydryl-*N*-(2-(2,6-diethylphenylimino)acenaphthylidene)-4-methylbenzenamines. The long branches of polymer which were linear acted as hard phase and highly branched structure which distributed among long chain acted as soft matrix. Thus, RHBLPE shows good elastomeric properties over a wide range of temperatures, with good elongation at break and elastic properties [85]. Copolymers of ethylene-1-octene were synthesized using living coordination polymerization. These polymers have good toughness and elasticity than random metallocene copolymers [86]. POEs which are semicrystalline in nature were synthesized in two steps. First, the allyl-terminated syndiotactic or isotactic PP macromonomers were synthesized using bis(phenoxyimine)titanium or chiral ansa-zirconocene catalysts. Later pyridyl-amido hafnium catalyst was used for copolymerization of macromonomers, ethylene, and alpha-olefin with control over composition and side-chain incorporation [87]. The unsymmetrical alpha-diimine nickel catalyst showed good properties in ethylene polymerization such as thermal stability, ability to generate high molecular weight PE, and high activity. These catalysts were used in the synthesis of PP, 1-hexene, 1-octene, and 1-decene polymers. These catalysts can incorporate polar monomers such as methyl 10-undecenoate and 10-undecene-1-ol. Thus, high-performance materials were made using these catalyst [88]. 1-octene was polymerized to yield semicrystalline, branched PE-like material, with higher molecular weight and narrow molecular weight distribution. Polymerization was carried out using Ni(II) α -diimine complex, in combination with various aluminum alkyls and monomer concentration. The results showed that polymer synthesized using Ni-1/MAO and Ni-1/MMAO had lower crystallinity because of the different distribution of polymer chains [89]. POEs are synthesized using the living aryl-naphthyl-alpha-diimine Ni(II) "sandwich" catalyst complex. This catalyst gives high crystalline hard blocks from 1-decene and low crystallinity soft blocks from ethylene [90]. Chain walking polymerization was used to fabricate random copolymers of 1-octene (OCT) with 1-decene (DE) and cyclopentene (CPE). The copolymers prepared behaved as polyolefin elastomers with varying mechanical properties due to variation in crystallinity, composition, and structure. The catalyst used was Ni(II) α -diimine complex [91]. Transparent POEs are prepared using a hard segment of polynorbornene and a soft segment of atactic PP, poly(ethylene-co-propylene), and poly(ethylene-co-1-hexene). The catalyst used was 1-MMAO/(2,6-di-*tert*-butyl-4-methylphenol) (BHT) system, where BHT used as a modifying agent of trialkylaluminum in MMAO system. The introduction of hydroxyl groups on polynorbornene (PNB-*b*-aPP-P(NB-co-U-OH)) improved its mechanical properties. These block copolymers had excellent toughness and thermal properties with elastic recovery [92]. Copolymerization of PP with higher alpha-olefins such as 1-octene, 1-dodecene, 1-hexadecene was done using pyridylamidohafnium catalyst was carried out. These copolymers have high molecular weight with narrow molecular weight distribution. With about 12 mol% of comonomer content, these polymers

had shown excellent ductile properties. With 1-hexadecene, copolymers had shown good elastic recovery and flexibility [93].

Composites of TPO

Incorporation of additives in organic polymer has found their various applications in the field of automobile, medical, electrical, and thermal. These materials deliver better properties than their individual components and neat polymers, such as enhancing mechanical, electrical, and thermal stability [94, 95]. Various fillers have been used in neat thermoplastic polyolefin elastomers or its blends to improve its processing and performance properties. Some of the important additives are discussed below.

Talc

TPOs based on PP and rubber are compounded with talc with varying the quantity of rubber and talc. It is reported that with the increase in talc content, it resulted in

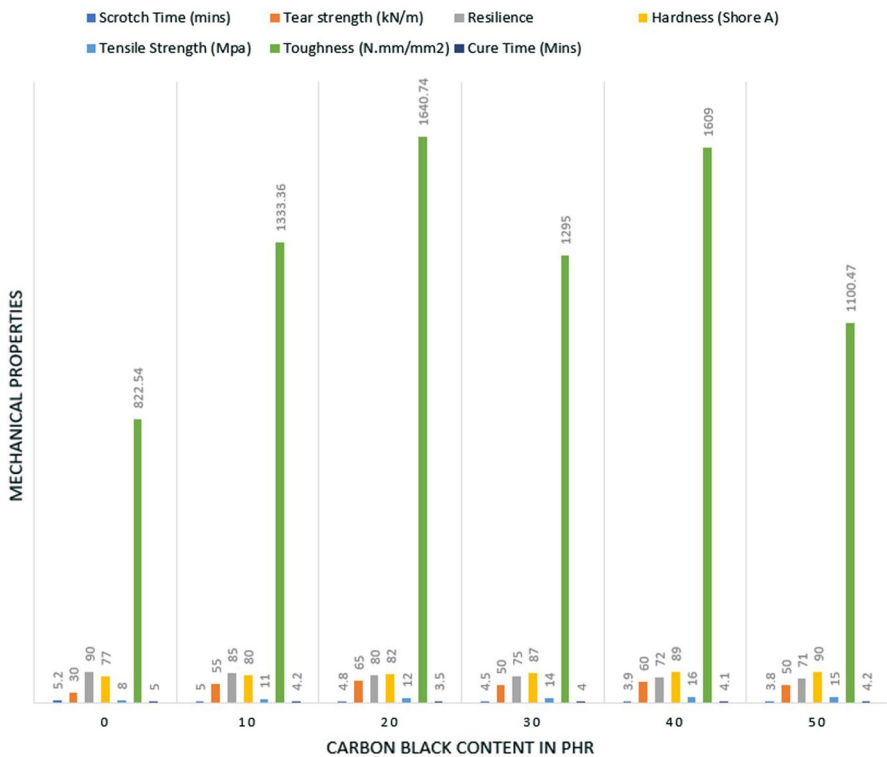


Fig. 5 Effect of carbon black (in phr) content on the mechanical properties of 80:20 NR/PP blends

higher modulus but lower strength and elongation at break at operating temperatures [96]. Later study also showed that, with the use of untreated talc it led to reaction with slip agents such as oleamide and erucamide used in composite and thus restricting them to reach to the surface and acting as hindrance in their functioning. Also it was noted that with the use of surface treated talc, it showed improved stiffness and hardness along with uniform dispersion of filler throughout the matrix for the composite [97]. Talc-filled rTPOs (reactor made TPO) composites were synthesized with various grades of talc with varying median diameter. It was seen that on incorporation of talc in higher levels, the elongation at break and impact strength decreased indicating brittle nature of composite. But with talc incorporation heat deflection temperature was found to increase [98].

Carbon black

The carbon black was incorporated in natural rubber/PP blend and it was reported that with increase in carbon black content, tensile properties improved. Various properties affected by carbon black content are given in the graph in Fig.5 [99]. Natural rubber/HDPE composites were prepared using carbon black. TEM analysis revealed that NR/HDPE was phase separated in blend structure and carbon black was incorporated dominantly in NR domains. This led to improved elastic properties and mechanical properties of the composite due to the presence of bound rubber. Bound rubber is formed when the surface of carbon black covered by a layer of rubber [100].

Organoclay

TPO/clay nanocomposites were studied containing 70% isotactic PP (iPP) and 30% EPDM and with very low loading of clay but found to have improved mechanical properties of composites [101]. The thermoplastic vulcanizates (TPV)/organoclay composites were also prepared successfully [102]. Later TPO/clay nanocomposites were prepared with MA-PP as compatibilizer. The tensile modulus and storage modulus of the composite were found to be improved [103]. Composites of maleated elastomer/PP and clay were synthesized by keeping the ratio of 70:30 constant. Elastomers used were EOC-g-MA and EPR-g-MA. EOC-g-MA/PP/clay composite showed improved ductility and stiffness and increased temperature for thermal degradation but reducing Young's modulus than EPR-g-MA composites [104]. Another study on clay nanocomposites was done by a melt compounding mixture of PP/POE (Ethylene-octene copolymer)/masterbatch. Masterbatch consisted of MA-PP and organically modified montmorillonite. The addition of elastomer increased the linear coefficient of thermal expansion but, it decreased with the addition of clay in both flow and transverse direction, among which reduction in the coefficient of thermal expansion in flow direction was greater than in transverse direction [105].

Silica

Polymer and silica composites are studied widely due to their improved mechanical properties along with reduced flammability and improved barrier properties [106, 107]. A study was done with PP, maleated PP, elastomer with SiO_2 as filler. Elastomers used were of two types: ethylene-octene copolymer and ethylene-propylene rubber. The use of maleated PP showed SiO_2 dispersed uniformly in the PP matrix. Optimum filler content reported was up to 5 wt% for PP/PP-g-MA/EOC blend which improved mechanical properties such as high elongation at break, yield stress, and young's modulus [108–110]. Elastomer rich TPO was synthesized using 70 wt% maleated styrene-ethylene-butadiene-styrene (SEBS-g-MA) and 30 wt% of PP with 1–5 wt% SiC nanoparticles (SiCp). It was reported that with SiCp addition, tensile stiffness and strength reduced but impact strength and fracture toughness were improved. Thermal resistance was found to be improved with SiCp nanoparticles [111]. A LDPE/EVA blend was chosen for studying the effect of nanofiller loaded into it, where pristine nanosilica and trimethoxyoctylsilane modified nanosilica were used as filler. Two different sequences were used for loading various components, and its effect was studied on mechanical properties of blends. It was revealed that the sequence of adding components affected properties of blend containing unmodified nanosilica and did not affect properties of modified nanosilica blend. With unmodified silica, at about 3 wt% loading, optimum mechanical properties reported were 4 MPa tensile strength and 520% elongation at break. This was explained as, at this loading level unmodified silica was dispersed in the elastomer phase. With silica modification, increasing in loading it showed improvement in mechanical properties. With about 5 wt% of loading, 5.3 MPa of tensile strength and 667% of elongation at break was obtained. This was related to formation of covalent bonding and good dispersion of nanosilica. Table 6 shows the effect of loading sequence on unmodified silica blends and loading level of modified silica blends [112].

Table 6 Few important mechanical properties of LDPE/EVA blends containing modified and unmodified silica [112]

Sr. No	Sample (with unmodified silica)	Tensile strength (MPa)	Elongation at break (%)
1	Virgin (C)	3.4	480
2	CS/1.5–2	3.8	478
3	CS/3–1	3.9	479
4	CS/3–2	4.0	520
5	CS/5–2	3.5	450
Sr. No	Sample (with modified silica)	Tensile Strength (MPa)	Elongation at break (%)
6	CmS/1.5–2	4.4	619
7	CmS/3–1	4.7	646
8	CmS/3–2	4.7	629
9	CmS/5–2	5.3	667

Magnesium-based filler

Fire retardant applications of polymer need to have high filler content but with high filler loading properties like tensile strength and elongation at break are severely affected. Thus, the use of co-continuous morphology was studied widely. It was reported that with PP/POE content 50:50, there existed co-continuous morphology along with magnesium hydroxide (MDH) content ranging from 55 to 60%. This highly filled composite shows good mechanical properties maintaining tensile strength and elongation at break. Hence, co-continuous morphology is in favor of tolerating high loading filler with retention of flexible properties [113]. PP/POE/MgO nanocomposites have shown enhanced tensile strength and modulus due to the synergistic effect of MgO and POE. It showed enhanced mechanical and electrical properties, and they have the potential to be used as recyclable insulation materials [114].

Fibers

The composite system of Kevlar ((poly(p-phenylene terephthalamide)) and santropene (POE) were studied. Untreated and treated Kevlar fibers are used to reinforce POE. It showed that using MA-g-PP as a compatibilizer and hydrolyzed Kevlar gave improved modulus, tensile strength, and elongation at break for the composite. The reaction between maleic anhydride groups on PP and free amine groups of Kevlar modified the surface of Kevlar thereby making the fiber nonpolar. These Kevlar fibers were able to co-crystallize with PP containing the POE matrix, leading to better stress distribution [115]. The study was done with polyethylene terephthalate (PET) fibers used as a reinforcing agent in TPO comprising PP matrix and EOC as an elastomer. Blend showed improved tensile and flexural behavior with an increase in PET fibers [116]. Composite containing PP matrix in TPOs along with PP homopolymer fiber reinforcement showed an increased processing window for the novel PP composite. The good interfacial tension between matrix and fiber was obtained. This PP composite had shown an increase in yield and tensile modulus [117].

Bio-composites

Kenaf-based bio-composites have Kenaf powder incorporated in recycled HDPE and natural rubber blends. It showed that with an increase in Kenaf content, a decrease in tensile strength, elongation at break was observed which was attributed to the agglomeration of fillers along with de-wetting of polymer and decrease in chain mobility with a rigid interface between fiber and matrix, respectively. With increased Kenaf powder stabilization torque was reported to be increased [118, 119]. Further modification in this study was made with MAPE when used as a compatibilizer. With Kenaf powder used along with MAPE it shown to improve tensile strength, elongation at break, and modulus. Also water absorption capacity of the

bio-composites decreased, thereby showing overall improvement [120, 121]. Later on, replacing Kenaf powder, Kenaf-based fibers were reported to be used with PP and different elastomers, namely EPDM and NR along with MA-PP as compatibilizer [122, 123]. Significant improvements in tensile strength were shown by PP/EPDM/MA-PP along with Kenaf fibers. This was attributed to the interaction of both matrix and Kenaf fiber [124]. Reinforcement of ethylene-octene copolymer with cereal straw fibers was done along with a cross-linking agent. These bio-composites regardless of straw type showed an increase in thermo-oxidative resistance [125]. Pineapple-filled SANTOPRENE composites were melt blended and tested. The test performed showed the results that tear resistance, hardness in longitudinal direction increased up to 15% fiber content, but elongation at break shown to have decreased [126].

Carbon nanotubes

Carbon nanotubes in TPO with compatibilizer were studied. Due to the low aspect ratio, higher filler inclusion in TPO was done which led to an increase in stiffness with decrease in toughness, also high filler loading led to poor processing and thus CNTs were studied and able to replace conventional fillers [127–129]. Composites of PP/EPDM with single-walled CNTs/compatibilizer ratio 1:2 was prepared and studied. A high level of toughness in TPO with 0.5% SWCNTs and 1% compatibilizer was obtained [130]. The latter modification was made using POE by synthesizing (POE)/multi-wall carbon nanotubes composites. These composites showed improved electrical and mechanical properties giving conductive elastomeric performance used for industrial applications [131].

Conductive fillers

Conductive polymers are always fascinating and drive attention, as they find applications in anticorrosive coatings, chemical and vapor and solution vapor-based sensors, electronic components and fuel cells [132–135]. A new and interesting study was made on conductive TPE-O of PP/EPDM blend and its composite using carbon black (CB), carbon nanotubes (CNT-CB), and its hybrid (CB-CNT). Percolation threshold and factors affecting it were studied for all these three fillers. It showed that percolation threshold for composite depends on filler aspect ratio and dispersion of filler. Carbon black when used as filler showed lower percolation threshold with sharp insulator-transition region, whereas carbon nanotubes and hybrid showed wider insulation-conductor region and pre-percolation conductivity [136]. Another important filler used for enhancement of mechanical and conductive properties of thermoplastic polyolefin elastomers (TPE-O) was graphene nanoplates (GNPs). They have excellent physical properties, which included high stiffness, large surface area, excellent thermal conductivity, and intrinsic carrier mobility [137, 138]. Thus, work was carried out incorporating GNPs in TPE-O containing PP/NR/LNR in ratio in 70:10:20 with varying ratio of GNPs. Various mechanical and

electrical properties were evaluated. Mechanical properties such as tensile strength, young's modulus, and impact strength increased till optimum filler loading and then decreased, whereas elongation at break was found to decrease with increase in filler loading. Electrical conductivity was found to increase with increase in filler loading. Figure 6 shows the trend in mechanical properties of blend with varying GNPs concentration [139].

Other fillers

CaCO_3 is also used as filler in TPOs. In PP/POE/ CaCO_3 ternary system, the addition of POE and CaCO_3 improved impact strength without lowering PP's stiffness and tensile strength [140, 141]. Graphene when used as filler in poly(ethylene-ter-1-hexene-terdivinylbenzene) (POE) in high quantity improved electrical and mechanical properties. These composites were used in lightweight EMI shielding applications [142]. In high-frequency application, composites based on Engage POE 8100, and $\text{Ni}_0.8\text{Zn}_0.2\text{Fe}_2\text{O}_4$ (NZO) are used. With increasing filler content composite showed an increase in permittivity and permeability [143]. Fly ash-based waste PE/reclaimed rubber composites were synthesized using a silane coupling agent. It showed improved tensile strength, flexural strength, flexural modulus, impact strength, and hardness of composites with addition of coupling agent [144]. Fly ash (FA)-filled ternary composite of PP/POE (Engage 8440)/PP-g-MA was prepared. FA content in samples varied from 0 phr to 20 phr with each sample comprising of 5 phr of PP-g-MA as compatibilizer with PP. It was seen that fly ash was dispersed well in PP matrix also study revealed that, between PP chains and fly ash particles, there exists hydrogen and covalent bonding due to compatibilizer used. Thermal stability of PP/POE fly ash composites was found to increase with fly ash content. Mechanical properties like tensile strength and flexural strength were found to increase with fly ash content [145]. Fire retardant TPO was made using

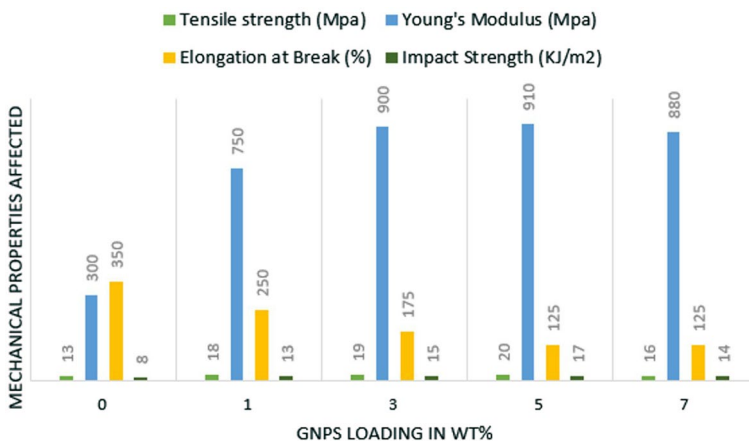


Fig. 6 Effect of graphene nanoplates content (in wt%) on NR/PP blends

ammonium dihydrogen phosphate (ADP) and starch (with quantity 100/60/20). It showed that in the presence of starch and ADP in ratio 1:3 the thermal stability of composite was increased [146]. PP/EPDM fire retardant composites were also synthesized using about 30–35% reactive melamine phosphate-pentaerythritol (MP/PER) [147]. Another study based on TPE-O available in market was with graphene nanoplates (GNPs) composites for exploring its various properties. They varied diameter of GNPs and its loading quantity and studied its effect. TPE-O used was Alcryn[®] 2265 UT. GNPs used were varied from 1–25 micrometer lateral diameter and 6–8 micrometer thickness. GNPs loading varied from 1 to 20 wt%. With increase in size of GNPs, stiffness of composites increased and smaller GNPs size lead to increase in strength of composites. Later it was revealed that, with further increase in GNPs than optimum level, mechanical properties were found to decrease due to agglomeration [148]. Some special fillers possessing antibacterial properties were also studied and incorporated into PP/POE (Versify 3401) blends. Ag-ZnO hierarchical particles were prepared using solutions of silver nitrate, zinc hexahydrate, and ammonium carbonate. The obtained powder was dried and compounded with PP/POE blends from 1–5 wt%. These composites were tested against *E. coli* and *S. aureus* and found successful. Thus, these composites can be used in sanitary, interiors, and other health-related applications [149].

Applications

The major advantage of using thermoplastic polyolefin elastomers is that it can be recycled as thermoplastics and has less curing time than thermoset rubbers [150]. These TPOs are modified with various additives finding its application in automobiles, insulation, etc. [7]. They are in-demand due to their design flexibility, styling, and low cost [151, 152]. Major applications of TPO are found in: automobile, wire, and cable, mechanical goods [153].

Automotive

Automobile applications include bumper covers, air dams, bodyside moldings, fender liners, and other applications involve conveyor belts [8]. In the automobiles sector, TPO faces a major drawback that the painted TPO parts are susceptible to damage known as comprehensive shear-induced delamination. Thus, an optimum combination of thermoplastic polyolefins and elastomer was made considering molecular weight of polyolefins, crystallinity, and elastomer content [154]. Intumescent materials made of TPO with HDPE as a hard phase and silicone rubber as a soft phase along with other additives are used for airplanes and motorcycles. They possess good mechanical properties and can be recycled and can be made from recycled materials [155]. Polyolefin elastomers (POE) can also be welded to TPO which is an important feature for them to be used in the automobile industry [156]. Foaming of TPO will be of great interest in the automobile industry as large parts with low thickness can be made, which depends on the particle size of the dispersed

phase in TPO [157]. A blend of recycled PP (rPP) and TPV was used as raw material for original equipment market (OEM) in Thailand and study showed that, on incorporating rPP in TPV, mechanical properties of blend like stress at break and flexural strength were found to increase [158].

Wires and cable

Earlier, cross-linked PE (XLPE) was used as insulation material for HDVC cables, due to its dielectric, mechanical, and thermal properties. But later some drawbacks were reported for XLPE [159]. Later TPE-O were used for insulation of cables [160–162]. PP/POE blend have the potential of forming recyclable HVDC cables for insulation applications. POE improves the flexibility of PP with excellent thermal properties, with breakdown strength slightly decreased and heterospace charge accumulation being enhanced [163]. POE/PE blends along with some nanocomposites are used in HVDC cables. These blends consist of PE/SEBS-MA/zinc oxide (ZnO), in which ZnO is dispersed in SEBS-MA or at the interface with SEBS-MA and PE. They reported in reduction of dielectric loss, thus used in HV insulation systems [164, 165]. Blending PP with POE (ethylene-octene copolymer) along with nano-silica found its application as the best material for HVDC cables [166]. TPO heat shrink tubing is used to protect wires from abrasion resistance as well as provide environmental protection. It protects wires from rain, dust, solvents, and holds it tight at the right place [18].

Medical grade

Medical grade TPO is used in orthopedic and dental applications. Medical grade states that a particular grade of TPO is subjected to various testing to be used as in medical applications. TPOs in the medical category are used for endoprosthetic applications [167]. Various medical devices are fabricated using thermoplastic elastomeric alpha-olefin which includes slips, staples, other fasteners, pins, screws, wound dressings, and drug delivery applications [168].

Roofing membranes

TPO roofing composition has polyolefins and rubbers blended together. This technology has given flexible roofing membrane without using plasticizers. As these TPO membranes do not contain halogenated compounds like chlorine, these are considered as “green” product technology. Reinforced TPO membranes are also studied and used commercially. They contain reinforcing scrim (fabric) between two TPO layers. This fabric can be glass scrim or polyester fabric. TPO is generally compounded with various additives such as UV stabilizers, fire retardants, and pigments which are used for high reflectance or for colored roofing membranes [169–176]. A study was made using various fillers along with NR/HDPE blend for roofing applications. In that work, six particulate fillers were incorporated in blend of NR/HDPE 20/80 ratio with varying concentration from 10 phr to 30 phr with interval

if 10 phr. They used six different fillers such as calcium carbonate, barium sulfate (BaSO_4), kaolin, talc, snobrite clay, and dolomite. Among those fillers used, calcium carbonate, talc, and kaolin were extenders while barium sulfate and dolomite were used as functional fillers. Talc, dolomite, and kaolin shown to reduce impact strength, whereas calcium carbonate, barium sulfate, and snobrite improved impact strength at optimum loading at 10 phr, 30 phr, and 20 phr, respectively. Among all fillers, BaSO_4 used showed highest improved properties such as tensile strength, tear strength, and hardness up to values of 17.8 MPa, 102.9 N/mm, and 58.5 shore D, respectively [177].

Other applications

TPO is widely used as encapsulating material for components which require special manufacturing technique, such materials include solenoids, sensors, transformer, motors, self-supporting coils, and electric components [18].

POE is now replacing conventional rubbers due to its processing and properties. POE is used in microwave applications. Ethylene-octene copolymer is ceramic filled

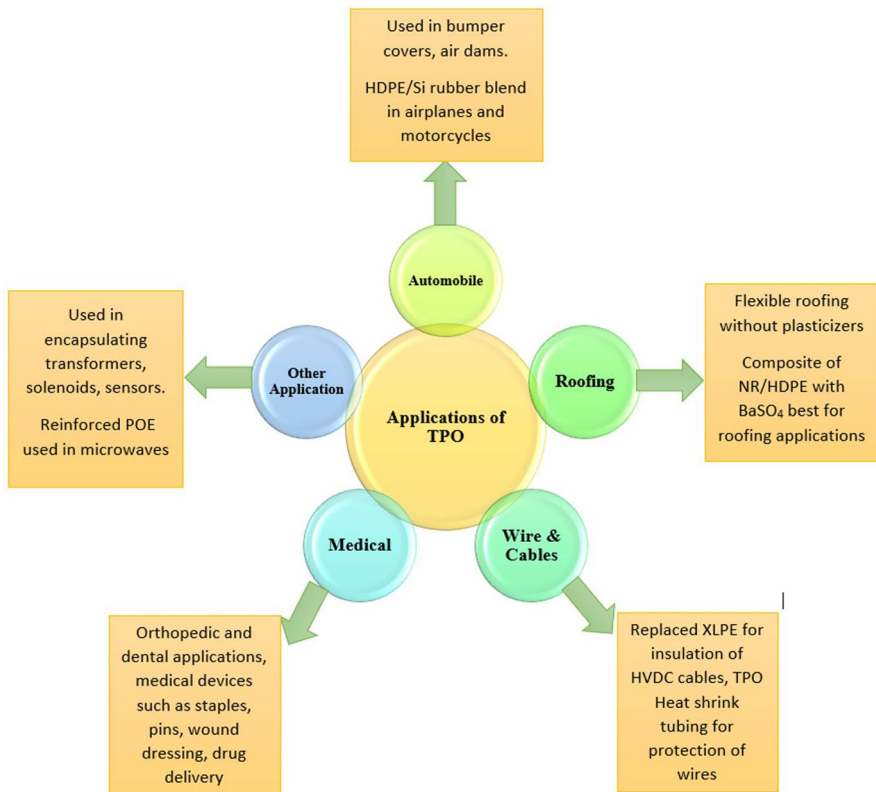


Fig. 7 Applications of TPO in various fields

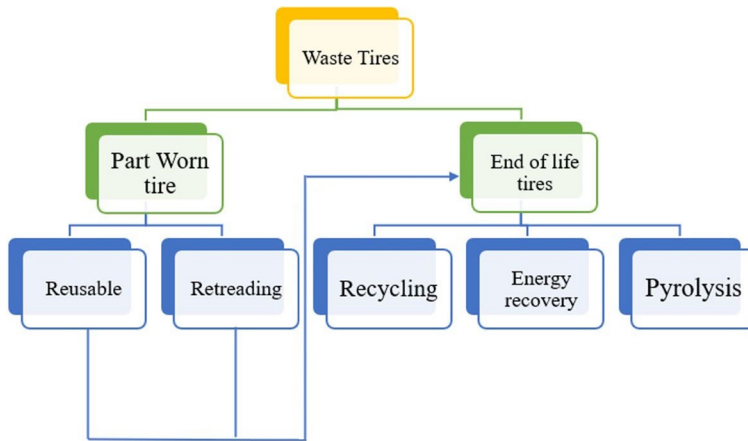


Fig. 8 Life cycle of waste tires

to improve its electrical properties making it capable to be used in microwave applications. EOC (POE) is, thus, reinforced with SrTiO_3 . SrTiO_3 is used because of high relative permittivity, low loss tangent, and has got mechanical properties [21].

POE is now used in radio frequency applications. POE is reinforced with SrTiO_3 (STO) and NiZn ferrite (NZO) ultrafine particles, such composites are used in capacitor–inductor integrating devices [178]. Films were prepared using polyolefin elastomer available in market Engage 8200 using extrusion and then electron beam (EB) irradiation. It showed that using EB irradiation, there found cross-linking structured among the chains resulting in increase in modulus and strain. A comparable strain to conventional polyurethane (PU) elastomer of about 1400% at 150 kGy was obtained [179]. Near-IR responsive shape memory POE composites with healing properties were synthesized and studied. Maleated POE was cross-linked using 3-amino-1,2,4-triazole (ATA) and compounded using octadecylamine modified graphene oxide (ODA-GO). It showed that with high loading of filler, its mechanical properties like tensile strength improved. Thus, 0.5 wt% of filler was considered optimum as sample had the highest healing efficiency and from 0.5 to 1.0 wt% of filler loading, sample recovered its original shape in three minutes [180]. Figure 7 gives overview of application of TPE-O/TPO.

Why TPE/TPE-O?

From above discussion and points, it can be seen that extensive research has been carried out on recycling of rubber waste around us, in which rubber tires have their major contribution. Also, these tires have various additives such as vulcanizing agents, oxidation resistance additives which make them resistant to extreme weather conditions [181, 182]. The accumulation of tires on the landfills leads to leaching out of these harmful additives into soil and then into ground waters [183]. The life cycle of tires after use is depicted in Fig. 8 [184]. Along with rubbers, plastics are also dumped and need to be recycled. Plastics like PP and PE need to be recycled

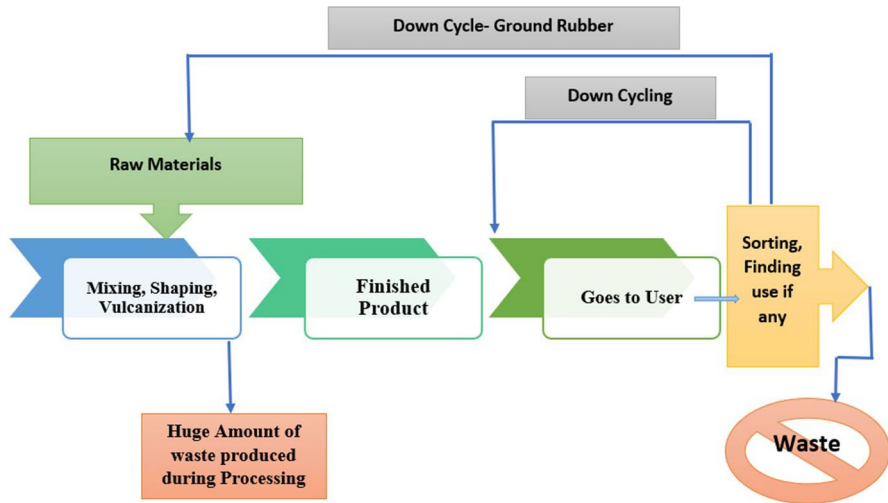


Fig. 9 Processing and recycling of conventional rubbers/elastomers

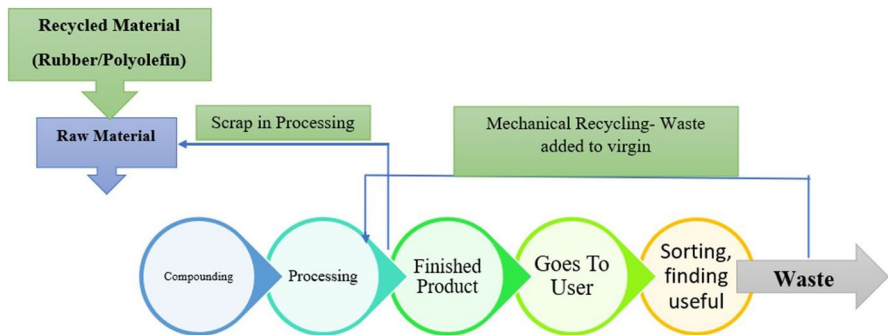


Fig. 10 Processing and recycling of thermoplastic polyolefin elastomers

and reused to reduce the load on environment due to land filling [185]. Yet major threat to environment are rubbers/elastomers, which are cured and shaped into desired shape required. Major source of this type of waste is automobile industry which comprises of components such as tires, weather strips, vibration isolators, and hoses [186]. Among rubbers/elastomers used in automobile industry, EPDM is most used rubber followed by natural rubber, and SBR. To recycle this used rubber, “Pan” process was used conventionally which gave low-quality reclaimed rubber as compared to virgin rubber. Thus, new process of continuous recycling was used which controlled selective bond cleavages in rubber particles, thereby resulting in molecular weight distribution and properties comparable to virgin rubber, and thus can be used for making new blends [186, 187]. Comparing solely POEs and conventional rubbers without blending it with polyolefins, it can be seen that POEs are synthesized using catalytic systems and do not undergo vulcanization process, and have

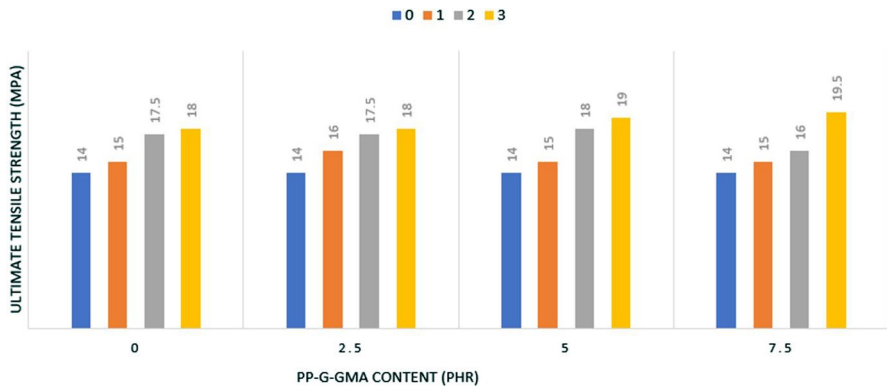


Fig. 11 Effect of reprocessing cycles (0 to 3) on tensile properties of PP/XNBR 50:50 with compatibilizer (PP-g-GMA in phr) [36]

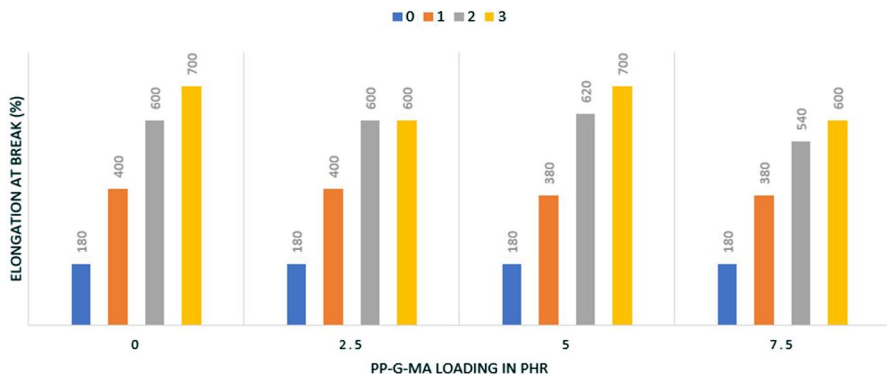


Fig. 12 Effect of reprocessing cycles (0 to 3) on elongation at break of PP/XNBR 50:50 with compatibilizer (PP-g-GMA in phr) [36]

got good elongation at break and can be compounded easily to improve mechanical properties in polyolefin blends [188]. Also, these POEs have got low level of unsaturation than conventional rubbers and hence have got good resistance against UV and heat. Having got good flexibility, they can be used without incorporation of plasticizers in interior of automobiles making them more sustainable than using conventional rubbers [189]. Now, comparing blends of elastomers/polyolefins (TPE-O) with conventional rubbers/elastomers it can be seen that TPOs were stable and exhibited excellent properties up to six cycles of recycling. Also, during processing of polymers lot of processing waste is discarded, in the case of TPOs, it can be easily reinstated along with virgin material but treated as waste when material is cross-linked or vulcanized as during processing of conventional rubbers. Figures 9 and 10 show the process of manufacturing and scrap recycling of conventional rubbers and TPE-O, respectively, depicting ease of processing and recycling for TPE-O. Also, it can be seen that high amount of rubber waste, special vulcanizing agents,

and numerous processing aids along with slow fabrication are some of the major drawbacks for thermoset rubbers. [190, 191]. A quantitative study on dynamically vulcanized PP/XNBR blend was carried out, stating that they can be subjected to processing for about three times, where it was depicted that ultimate tensile strength remained consistent throughout the process and elongation at break was decreased slightly. This study proved that reprocessing of TPE-O based on PP/XNBR can be done with retention of original properties such as tensile strength and elongation at break, which are depicted in Fig. 11 and Fig. 12, respectively [36]. Another study using ground rubber tire, EPDM and acrylic acid modified HDPE showed that these blends (TPE-O) can be reprocessed and their properties are retained well after reprocessing. After three reprocessing cycles, not major change was reported in tensile strength but elongation at break was found to increase. Thus, when such materials used can be recycled and reprocessed into products of value ultimately it reduces load on environment. Along with rubbers, recycled thermoplastics can also be used in making TPE-Os [55]. With micron-size tire rubber particles, recycled TPOs from automobile industry and maleinized polybutadiene (as compatibilizer) were used for making value added products. It was concluded that balance between addition of rubber particles to TPO has to be established well to obtain optimum properties [192]. Looking at the study done by various researchers above, it can be safely assumed that recycling of TPE-O as individual material can be done easily than vulcanized/cross-linked rubbers. Replacing conventional rubbers completely cannot be done at this point for all applications, but sustainable replacement can be used at places where ever made possible, which can help reduce carbon footprint.

Conclusion

A thorough study on thermoplastic polyolefin elastomers (TPO) is made in this paper. It is reported that the mechanical blending of thermoplastic olefins with rubbers improved impact strength, ductility, and other properties of the virgin olefins and rubbers. Though mechanical blending is an economical process and has a wide scope but has the limitation of miscibility of two phases. Thus, compatibilizers are used in blending to reduce the interfacial tension between blended phases. Copolymerization of olefinic monomers was done using catalytic systems to obtain better elastomeric properties. Vulcanized waste rubbers can be devulcanized and incorporated in the polyolefin matrix to improve properties as well as reduce stress on the environment. TPOs are compounded with different types of fillers such as organo-clay, talc, MgO, $\text{Mg}(\text{OH})_2$, and silica nanoparticles. Compounding of TPO was studied for improving properties such as impact strength, elongation at break, tear resistance, and toughness. Synthetic and bio-composites of TPOs are of tremendous importance in a microwave, electrical insulation, fire retardants plastics, etc.

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